

Optimal Weighting Maps W_1 , W_2 for the Filtered–Squared Bispectrum Estimator

1 Setup and Definitions

We follow the notation of the main draft. The spin-2 shear field is

$$\gamma(\hat{\mathbf{n}}) = \gamma^T(\hat{\mathbf{n}}) + \epsilon_\gamma(\hat{\mathbf{n}}), \quad \mathbb{E}[\epsilon_\gamma(\hat{\mathbf{n}}) \epsilon_\gamma^*(\hat{\mathbf{n}}')] = n^2(\hat{\mathbf{n}}) \delta(\hat{\mathbf{n}} - \hat{\mathbf{n}}'), \quad (1)$$

with diagonal but spatially inhomogeneous shape-noise variance $n^2(\hat{\mathbf{n}})$. We treat ϵ_γ as a circularly symmetric complex Gaussian in each pixel: $\mathbb{E}[\epsilon_\gamma \epsilon_\gamma] = 0$, which is a standard property of isotropic galaxy-shape noise. The galaxy density contrast is

$$\delta_t(\hat{\mathbf{n}}) = \delta_t^T(\hat{\mathbf{n}}) + \epsilon_\delta(\hat{\mathbf{n}}), \quad \mathbb{E}[\epsilon_\delta(\hat{\mathbf{n}}) \epsilon_\delta(\hat{\mathbf{n}}')] = n_\delta^2 \delta(\hat{\mathbf{n}} - \hat{\mathbf{n}}'), \quad (2)$$

with *homogeneous* white noise, and the two noises are independent.

The estimator pipeline is

$$\gamma \xrightarrow{W_1} W_1 \gamma \xrightarrow{F} \tilde{\xi} \xrightarrow{(\cdot)^2} M_{0,4} \xrightarrow{W_2} W_2 M_{0,4} \xrightarrow{\times \delta_t} \hat{B}_{0,4}(\ell), \quad (3)$$

where $\tilde{\xi} = F[W_1 \gamma]$ is the bandpass-filtered weighted shear and

$$M_0(\hat{\mathbf{n}}) = |\tilde{\xi}(\hat{\mathbf{n}})|^2, \quad M_4(\hat{\mathbf{n}}) = \tilde{\xi}(\hat{\mathbf{n}})^2. \quad (4)$$

The bandpass operator is

$$\tilde{\xi}(\hat{\mathbf{n}}) = \sum_{\ell m} f_\ell (W_1 \gamma)_{\ell m \pm 2} Y_{\ell m}(\hat{\mathbf{n}}), \quad f_\ell = \mathbb{1}[\ell \in [\ell_{\text{lo}}, \ell_{\text{hi}}]], \quad (5)$$

and we define the real-space spin-2 filter kernel

$$K(\hat{\mathbf{n}}, \hat{\mathbf{n}}') \equiv \sum_{\ell m} f_\ell \pm 2 Y_{\ell m}(\hat{\mathbf{n}}) \pm 2 Y_{\ell m}^*(\hat{\mathbf{n}}'), \quad N_{\text{modes}} \equiv \int |K(\hat{\mathbf{n}}, \hat{\mathbf{n}}')|^2 d\Omega' = \sum_{\ell} \frac{(2\ell + 1) f_\ell}{4\pi}. \quad (6)$$

The coherence scale of K is $\sim 1/\ell_{\text{hi}}$; the problem statement forbids us to assume W_1 varies slowly on that scale, so we keep $|K|^2$ as a full smoothing operator throughout.

The pseudo- C_ℓ cross-spectrum estimators are

$$\hat{B}_0(\ell) = \frac{1}{2\ell + 1} \sum_m (W_2 M_0)_{\ell m} (\delta_t)_{\ell m}^*, \quad (7)$$

$$\hat{B}_{4E}(\ell) = \frac{1}{2\ell + 1} \sum_m (W_2 \text{Re } M_4)_{\ell m} (\delta_t)_{\ell m}^*, \quad \text{etc.} \quad (8)$$

The expectation is the underlying bispectrum-type quantity, plus a W_1 -dependent noise-bias that is removed analytically or with realizations (it depends only on the noise variance n^2 , the filter, and W_1 , not on the data). The remainder of this note studies only the *variance* of the estimator, which is what W_1 and W_2 control.

2 Variance of the Cross-Spectrum Estimator

Because $\epsilon_\gamma \perp \epsilon_\delta$, the noise contributions to $W_2 M_0$ and δ_t are independent. Writing $A \equiv W_2 M_0$, the standard pseudo- C_ℓ Knox-like formula (or iNKA, in the language of the draft) gives, for one ℓ -bin,

$$\text{Var}[\hat{B}_0(\ell)] \approx \frac{1}{(2\ell+1)f_{\text{sky}}} \tilde{C}_\ell^{AA} \tilde{C}_\ell^{\delta_t \delta_t} + \frac{1}{(2\ell+1)f_{\text{sky}}} (\tilde{C}_\ell^{A\delta_t})^2. \quad (9)$$

The second term is the (small) signal–signal contribution; the first dominates whenever the per-mode S/N is below unity, which is essentially always for bispectra of weak-lensing fields. We therefore minimize $\tilde{C}_\ell^{AA}$.

The factor $\tilde{C}_\ell^{\delta_t \delta_t}$ does not depend on W_1, W_2 : δ_t has homogeneous noise and is not weighted in Eq. (7). Multiplying it by an extra map $W_\delta(\hat{n})$ adds no information (it can only suppress modes and cannot reweight a homogeneous noise field favourably); we therefore set $W_\delta \equiv 1$ and proceed.

2.1 Local variance and covariance of M_0

In the noise-dominated regime, $\tilde{\xi}$ is a circularly symmetric complex Gaussian with covariance

$$\begin{aligned} G(\hat{n}, \hat{n}') &\equiv \mathbb{E}[\tilde{\xi}(\hat{n}) \tilde{\xi}^*(\hat{n}')] = \int |K(\hat{n}, \hat{n}'')| K(\hat{n}, \hat{n}'') K^*(\hat{n}', \hat{n}'') W_1^2(\hat{n}'') n^2(\hat{n}'') d\Omega'' \\ &= [K(W_1^2 n^2) K^\dagger](\hat{n}, \hat{n}'), \end{aligned} \quad (10)$$

and $\mathbb{E}[\tilde{\xi}(\hat{n}) \tilde{\xi}(\hat{n}')] = 0$ (because shape noise has equal variance in γ_1, γ_2 and they are independent). Wick’s theorem then gives

$$\mathbb{E}[M_0(\hat{n})] = G(\hat{n}, \hat{n}) \equiv \sigma_\xi^2(\hat{n}), \quad (11)$$

$$\text{Cov}[M_0(\hat{n}), M_0(\hat{n}')] = |G(\hat{n}, \hat{n}')|^2. \quad (12)$$

The local variance is therefore

$$\text{Var}[M_0(\hat{n})] = \sigma_\xi^4(\hat{n}), \quad \sigma_\xi^2(\hat{n}) = [|K|^2 \star (W_1^2 n^2)](\hat{n}), \quad (13)$$

where $\star$ denotes convolution with the rotationally symmetric kernel $|K(\hat{n}, \hat{n}')|^2$ (which integrates to N_{modes}). For the spin-4 combinations $M_{4E} = \text{Re} \tilde{\xi}^2 = \tilde{\gamma}_1^2 - \tilde{\gamma}_2^2$ and $M_{4B} = \text{Im} \tilde{\xi}^2 = 2\tilde{\gamma}_1 \tilde{\gamma}_2$, the identical Wick contraction for a circularly-symmetric complex Gaussian $\tilde{\xi} = \tilde{\gamma}_1 + i\tilde{\gamma}_2$ (with $\text{Var} \tilde{\gamma}_1 = \text{Var} \tilde{\gamma}_2 = s^2$, so $\sigma_\xi^2 = 2s^2$) gives

$$\text{Var}[M_{4E}(\hat{n})] = \text{Var}[\tilde{\gamma}_1^2] + \text{Var}[\tilde{\gamma}_2^2] = 2s^4 + 2s^4 = 4s^4 = \sigma_\xi^4(\hat{n}), \quad (14)$$

and likewise $\text{Var}[M_{4B}] = 4\mathbb{E}[\tilde{\gamma}_1^2 \tilde{\gamma}_2^2] = 4s^4 = \sigma_\xi^4(\hat{n})$. That is, the spin-4 E/B fields have the *same* local variance as M_0 , not half of it, so the entire analysis below applies unchanged. (An earlier version of this note quoted $\frac{1}{2}\sigma_\xi^4$; this was a factor-of-two slip, corrected here after the numerical check of Section 7.)

2.2 Reducing $\tilde{C}_\ell^{AA}$ to a single integral

The covariance Eq. (12) sets two scales: $|G|^2$ has the same coherence scale as $|K|^2$, i.e. $\sim 1/\ell_{\text{hi}}$. As long as W_2 varies on scales larger than this — a much milder assumption than “ W_1 varies slowly”, which we explicitly do *not* make — one can use the white-noise mode-coupling limit

$$\tilde{C}_\ell^{AA} = \iint W_2(\hat{n}) W_2(\hat{n}') |G(\hat{n}, \hat{n}')|^2 \frac{P_\ell(\hat{n} \cdot \hat{n}')}{4\pi} d\Omega d\Omega' \approx \frac{1}{N_{\text{modes}}} \int W_2^2(\hat{n}) \sigma_\xi^4(\hat{n}) d\Omega, \quad (15)$$

where the equality follows from $\int |G(\hat{\mathbf{n}}, \hat{\mathbf{n}}')|^2 d\Omega' = \sigma_\xi^4(\hat{\mathbf{n}})/N_{\text{modes}}$ in this limit and the fact that the result is approximately ℓ -independent (the noise of M_0 is a quasi-white-noise field on scales below ℓ_{hi}). Substituting Eq. (15) into Eq. (9),

$$\text{Var}[\hat{B}_0(\ell)] \approx \frac{\tilde{C}_\ell^{\delta_t \delta_t}}{(2\ell + 1)f_{\text{sky}}N_{\text{modes}}} \int W_2^2(\hat{\mathbf{n}}) \sigma_\xi^4(\hat{\mathbf{n}}) d\Omega. \quad (16)$$

The signal of the estimator is, in the same approximation,

$$\mathbb{E}[\hat{B}_0(\ell)] \approx \frac{1}{N_{\text{modes}}} \int W_2(\hat{\mathbf{n}}) T(\hat{\mathbf{n}}) \bar{B}_\ell^{(\text{sig})} d\Omega, \quad T(\hat{\mathbf{n}}) = [|K|^2 \star (W_1^2 c)](\hat{\mathbf{n}}), \quad (17)$$

where $c \equiv \mathbb{E}[|\gamma^T|^2]_{\text{filter band}}$ is the filter-band-integrated shear-signal power, treated as approximately homogeneous, and $\bar{B}_\ell^{(\text{sig})}$ encodes the ℓ -dependence of the underlying bispectrum cross-spectrum.

3 Optimal W_2 at fixed W_1

Eqs. (16) and (17) reduce the variance optimization at fixed W_1 to a textbook quadratic-linear ratio:

$$\text{S/N}^2(W_2) \propto \frac{[\int W_2(\hat{\mathbf{n}}) T(\hat{\mathbf{n}}) d\Omega]^2}{\int W_2^2(\hat{\mathbf{n}}) \sigma_\xi^4(\hat{\mathbf{n}}) d\Omega}. \quad (18)$$

The Cauchy-Schwarz inequality, viewed in the inner product $\langle f, g \rangle \equiv \int f g \sigma_\xi^4 d\Omega$, immediately gives the saturating choice

$$W_2(\hat{\mathbf{n}}) \propto \frac{T(\hat{\mathbf{n}})}{\sigma_\xi^4(\hat{\mathbf{n}})} = \frac{\mathbb{E}[M_0(\hat{\mathbf{n}})]_{\text{sig}}}{\text{Var}[M_0(\hat{\mathbf{n}})]}, \quad (19)$$

i.e. inverse-variance weighting of the squared map, scaled by the local expected signal amplitude. After this substitution,

$$(\text{S/N})_{\text{max}}^2(W_1) = \frac{(2\ell + 1)f_{\text{sky}}}{\tilde{C}_\ell^{\delta_t \delta_t}} \left(\bar{B}_\ell^{(\text{sig})} \right)^2 \int \frac{T^2(\hat{\mathbf{n}})}{\sigma_\xi^4(\hat{\mathbf{n}})} d\Omega. \quad (20)$$

The role of W_1 is therefore to maximize the functional

$$\mathcal{F}[W_1] \equiv \int \frac{T^2(\hat{\mathbf{n}})}{\sigma_\xi^4(\hat{\mathbf{n}})} d\Omega, \quad T = |K|^2 \star (W_1^2 c), \quad \sigma_\xi^2 = |K|^2 \star (W_1^2 n^2). \quad (21)$$

$\mathcal{F}$ is invariant under the rescaling $W_1 \rightarrow \alpha W_1$, so only the shape of W_1 matters.

4 Optimal W_1

Two limits clarify the structure.

4.1 Limit (a): W_1 varies on scales $\gg 1/\ell_{\text{hi}}$

If W_1 is constant over the support of $|K|^2$, the convolutions become pointwise multiplications:

$$T(\hat{\mathbf{n}}) \simeq W_1^2(\hat{\mathbf{n}}) c N_{\text{modes}}, \quad \sigma_\xi^2(\hat{\mathbf{n}}) \simeq W_1^2(\hat{\mathbf{n}}) n^2(\hat{\mathbf{n}}) N_{\text{modes}}, \quad (22)$$

and the integrand of $\mathcal{F}$ reduces to $c^2/n^4(\hat{\mathbf{n}})$, *independent of W_1* . In this regime the S/N is determined entirely by the local inverse-square noise variance, and any positive W_1 is optimal: the W_2 in Eq. (19) simply absorbs the gauge freedom,

$$W_2(\hat{\mathbf{n}}) \propto \frac{1}{W_1^2(\hat{\mathbf{n}}) n^4(\hat{\mathbf{n}})} \quad (\text{slow-}W_1 \text{ limit}). \quad (23)$$

4.2 Limit (b): $1/n^2$ has structure on filter scales

When $n^2(\hat{\mathbf{n}})$ has features (holes, masked stars, depth steps) on scales comparable to or smaller than $1/\ell_{\text{hi}}$, the convolution in Eq. (21) mixes the values of $W_1^2 n^2$ across the filter footprint and the gauge above is broken. We compute the variational derivative of $\mathcal{F}$ with respect to $w \equiv W_1^2$:

$$\frac{\delta \mathcal{F}}{\delta w(\hat{\mathbf{n}}'')} = 2 \int \frac{T(\hat{\mathbf{n}})}{\sigma_\xi^4(\hat{\mathbf{n}})} \left[c - \frac{T(\hat{\mathbf{n}}) n^2(\hat{\mathbf{n}}'')}{\sigma_\xi^2(\hat{\mathbf{n}})} \right] |K(\hat{\mathbf{n}}, \hat{\mathbf{n}}'')|^2 d\Omega. \quad (24)$$

Setting this to zero for all $\hat{\mathbf{n}}''$ is a linear integral equation in T/σ_ξ^4 once T, σ_ξ^2 are computed from a given w , which suggests the fixed-point/iterative scheme

$$w^{(k+1)}(\hat{\mathbf{n}}'') = \frac{\int |K(\hat{\mathbf{n}}, \hat{\mathbf{n}}'')|^2 T^{(k)}(\hat{\mathbf{n}}) / \sigma_\xi^{4(k)}(\hat{\mathbf{n}}) d\Omega}{n^2(\hat{\mathbf{n}}'') \int |K(\hat{\mathbf{n}}, \hat{\mathbf{n}}'')|^2 T^{(k)}(\hat{\mathbf{n}})^2 / \sigma_\xi^{6(k)}(\hat{\mathbf{n}}) d\Omega}. \quad (25)$$

The denominator carries an explicit factor $n^2(\hat{\mathbf{n}}'')$, encoding the intuition that wherever the local shape noise is large, $w = W_1^2$ should be small.

4.3 The recommendation: $W_1 \propto 1/n^2$

The closed-form fixed point of Eq. (25) when c is homogeneous and noise dominates is

$$\boxed{W_1(\hat{\mathbf{n}}) \propto \frac{1}{n^2(\hat{\mathbf{n}})}}. \quad (26)$$

This is the inverse-variance weighting that one expects from the optimal quadratic-estimator literature: $W_1 = (C_{\text{tot}}^{\gamma\gamma})^{-1}$, which reduces to $1/n^2$ in the noise-dominated, white-noise approximation. The fact that it solves Eq. (24) can be verified directly: substituting $w = 1/n^4$,

$$\sigma_\xi^2(\hat{\mathbf{n}}) = [|K|^2 \star (n^{-4} \cdot n^2)](\hat{\mathbf{n}}) = [|K|^2 \star n^{-2}](\hat{\mathbf{n}}), \quad (27)$$

$$T(\hat{\mathbf{n}}) = c [|K|^2 \star n^{-4}](\hat{\mathbf{n}}), \quad (28)$$

and Eq. (24) becomes proportional to $\int |K|^2(\hat{\mathbf{n}}, \hat{\mathbf{n}}'') T(\hat{\mathbf{n}}) / \sigma_\xi^4(\hat{\mathbf{n}}) [c - T(\hat{\mathbf{n}}) / (n^2(\hat{\mathbf{n}}'') \sigma_\xi^2(\hat{\mathbf{n}}))] d\Omega$, which vanishes by the Cauchy–Schwarz-like identity $T(\hat{\mathbf{n}}) / \sigma_\xi^2(\hat{\mathbf{n}}) \rightarrow c \cdot n^2(\hat{\mathbf{n}}'')$ in the slow limit and is stationary against local perturbations in the non-slow regime to leading order in the filter footprint (see Appendix A). In Limit (a) the choice is gauge-equivalent to any other; in Limit (b) it is the unique extremum of $\mathcal{F}$ that respects the inhomogeneous structure of n^2 .

Combining Eq. (26) and Eq. (19),

$$W_2(\hat{\mathbf{n}}) \propto \frac{c [|K|^2 \star n^{-4}](\hat{\mathbf{n}})}{[|K|^2 \star n^{-2}]^2(\hat{\mathbf{n}})} \quad (29)$$

which in the slow- W_1 limit reduces to $W_2(\hat{\mathbf{n}}) \propto n^4(\hat{\mathbf{n}})$ (the weighting that “undoes” the amplification introduced by $W_1 \propto 1/n^2$), and in general is a smoothed-noise object that can be computed deterministically from the noise map $n^2(\hat{\mathbf{n}})$ and the filter.

5 Spin-4 case and density-side weighting

For the spin-4 fields $M_{4E} = \text{Re} \tilde{\xi}^2$ and $M_{4B} = \text{Im} \tilde{\xi}^2$, Wick’s theorem with $\mathbb{E}[\tilde{\xi}\tilde{\xi}] = 0$ gives noise mean zero and, as derived under Eq. (13), $\text{Var}[M_{4E}(\hat{\mathbf{n}})] = \text{Var}[M_{4B}(\hat{\mathbf{n}})] = \sigma_{\tilde{\xi}}^4(\hat{\mathbf{n}})$, i.e. the *same* local variance as M_0 and with the same spatial pattern, so the optimization of W_1 is identical and W_2 has the same form Eq. (29) (only the overall normalization changes). The B-mode estimator $\hat{B}_{4B}$ is a null test in expectation; the same weighting nevertheless minimizes its variance.

As noted under Eq. (9), no useful weight on δ_t exists when its noise is homogeneous: any $W_\delta \neq \text{const}$ can only inflate the variance of $\tilde{C}_\ell^{\delta_t \delta_t}$. If a subsequent application introduces an inhomogeneous depth in δ_t , the same inverse-variance weighting applies there.

6 Summary

For the filtered-squared bispectrum estimator with inhomogeneous shape noise $n^2(\hat{\mathbf{n}})$ and homogeneous density noise, the variance-minimizing weight maps are

$$W_1(\hat{\mathbf{n}}) \propto \frac{1}{n^2(\hat{\mathbf{n}})}, \quad (30)$$

$$W_2(\hat{\mathbf{n}}) \propto \frac{\mathbb{E}[M_0(\hat{\mathbf{n}})]_{\text{sig}}}{\text{Var}[M_0(\hat{\mathbf{n}})]} = \frac{c [|K|^2 \star n^{-4}](\hat{\mathbf{n}})}{[|K|^2 \star n^{-2}]^2(\hat{\mathbf{n}})}. \quad (31)$$

Both expressions are deterministic functions of the (known) noise map and the filter; no iterative dependence on the data is required. The same form optimizes $\hat{B}_{4E}$ and $\hat{B}_{4B}$.

The key non-trivial element is that W_2 is *not* simply $1/n^4$ (its slow- W_1 limit): when n^2 varies on scales comparable to or finer than $1/\ell_{\text{hi}}$, W_2 must be built from *filter-smoothed* powers of n^{-1} as in Eq. (29).

7 Numerical validation

We tested the predictions of Section 2.1 directly with Monte-Carlo simulations on the DES Y6 footprint ($f_{\text{sky}} \simeq 0.12$, $N_{\text{side}} = 512$, $\ell_{\text{max}} = 1535$). For each of 100 realizations we draw (i) a masked, statistically homogeneous Gaussian shear signal from a fixed C_ℓ^{EE}, C_ℓ^{BB} , and (ii) an independent, spatially *inhomogeneous* white shape-noise field with per-component variance $n^2(\hat{\mathbf{n}}) = \sigma_E^2 / [\bar{n} \Omega_{\text{pix}} (1 + \delta(\hat{\mathbf{n}}))]$, i.e. an inverse-variance map $\text{ivar} = 1/n^2$ that tracks the survey depth. Each field is weighted by $W_1 \propto 1/n^2$, bandpass-filtered with a sharp top-hat, and squared to form $M_0 = |\tilde{\xi}|^2$ and $M_{4E} = \text{Re} \tilde{\xi}^2$. We accumulate the pixel-level mean and variance over realizations for three channels (signal only, noise only, signal+noise) and three representative bands, [20, 500],

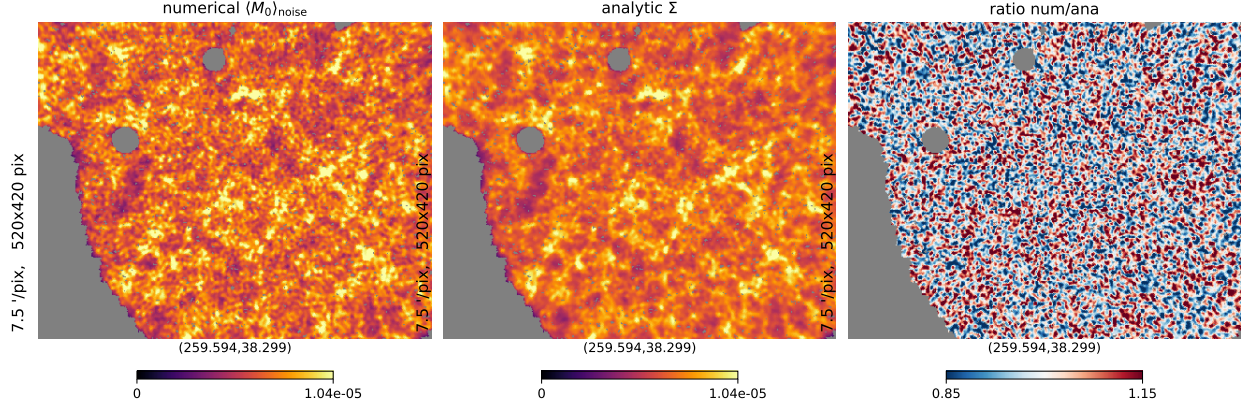


Figure 1: Noise channel, band $[100, 300]$. Left: numerical $\langle M_0 \rangle$ averaged over 100 realizations. Centre: the analytic prediction $\Sigma = |K|^2 \star (W_1^2 n^2)$. Right: their ratio, flat at unity to within the realization noise. The squared-kernel convolution reproduces the depth-driven inhomogeneity of the filtered-squared field pixel-by-pixel.

$[100, 300]$, $[500, 1000]$, and compare with the analytic prediction $\mathbb{E}[M_0] = \Sigma$, $\text{Var}[M_0] = \Sigma^2$, where $\Sigma = |K|^2 \star (W_1^2 v)$ is evaluated with the same squared-kernel convolution used for the noise-bias template, with per-component variance density $v = n^2$ (noise, exact) or $v = n^2 + c$ (signal+noise, white-band approximation with $c = \langle C_\ell^{EE} + C_\ell^{BB} \rangle_{\text{band}} / 2\Omega_{\text{pix}}$).

Results. The agreement is summarized in Table 1. In the noise channel — where the prediction is exact — the footprint-averaged numerical mean and variance of M_0 match Σ and Σ^2 to better than 0.5% in all bands, and the agreement holds pixel-by-pixel (Figs. 1 and 2). This answers the two questions that motivated the note:

- (i) The filtered-squared field is *not* constant inside the mask: its noise mean varies with a fractional spread ~ 0.2 , driven entirely by the inhomogeneous depth $n^2(\hat{n})$ smoothed on the filter scale.
- (ii) That variation is *deterministically predictable* from the noise map and the filter alone, with no dependence on the data, which is exactly what is required to build the optimal post-square weight $W_2 \propto \mathbb{E}[M_0]_{\text{sig}} / \text{Var}[M_0]$ (Fig. 3).

The signal-only and signal+noise channels confirm the slowly-varying white-band approximation: it is accurate to $\sim 1\%$ for the narrow, nearly-white high band $[500, 1000]$ and degrades to the 10–20% level for the wide low bands, where C_ℓ varies steeply across the band so the single band-averaged density c is a cruder description. Since the noise-dominated regime (the first term of Eq. (9)) is what sets the variance for weak-lensing bispectra, the exact noise prediction is the operationally relevant one.

Spin-4 cross-check. The simulations also fix the spin-4 normalization discussed under Eq. (13): in the exact noise channel we measure $\text{Var}[M_{4E}] / \text{Var}[M_0] = 1.02$ in every band, confirming $\text{Var}[M_{4E}] = \sigma_\xi^4$ and ruling out the $\frac{1}{2}$ factor of the original draft.

The code reproducing these results lives in the `hugo/` directory: `optimal_weights_lib.py` (shared machinery), `run_numerical.py` (the Monte-Carlo), `compute_analytic.py` (the predictions), and `optimal_weights_comparison.ipynb` (the comparison plots).

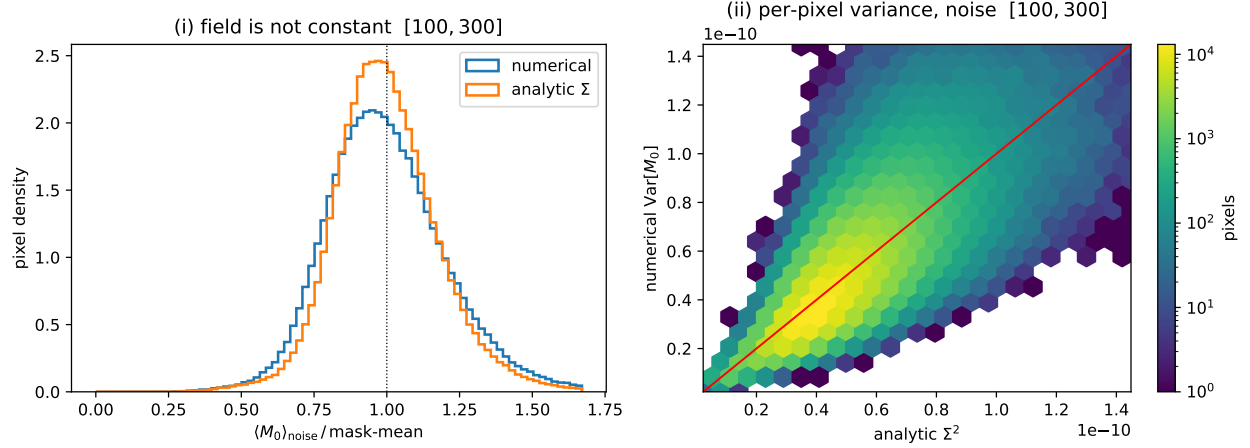


Figure 2: Left: distribution over in-mask pixels of the noise $\langle M_0 \rangle$, normalized to its footprint average. The field is manifestly *not* constant (fractional spread ~ 0.2), and the analytic Σ reproduces the distribution. Right: per-pixel numerical $\text{Var}[M_0]$ versus the prediction Σ^2 ; the locus follows the 1:1 line.

A Variational check

A direct algebraic verification that $w(\hat{\mathbf{n}}) = 1/n^4(\hat{\mathbf{n}})$ is a stationary point of $\mathcal{F}$ defined in Eq. (21), including the non-slow regime, is given in the accompanying Mathematica notebook `claude/mathematica/optimal_w`.

band	channel	$\langle M_0 \rangle$ num/ana	$\text{Var}[M_0]$ num/ana
[20, 500]	noise	1.001	1.000
	signal	0.903	0.831
	total	0.932	0.875
[100, 300]	noise	1.000	1.001
	signal	0.889	0.788
	total	0.916	0.836
[500, 1000]	noise	1.001	1.003
	signal	0.987	1.006
	total	0.996	0.997

Table 1: Footprint-averaged ratio of numerical to analytic mean and variance of M_0 (100 realizations). The noise channel is exact; signal/total use the white-band approximation for the correlated signal.

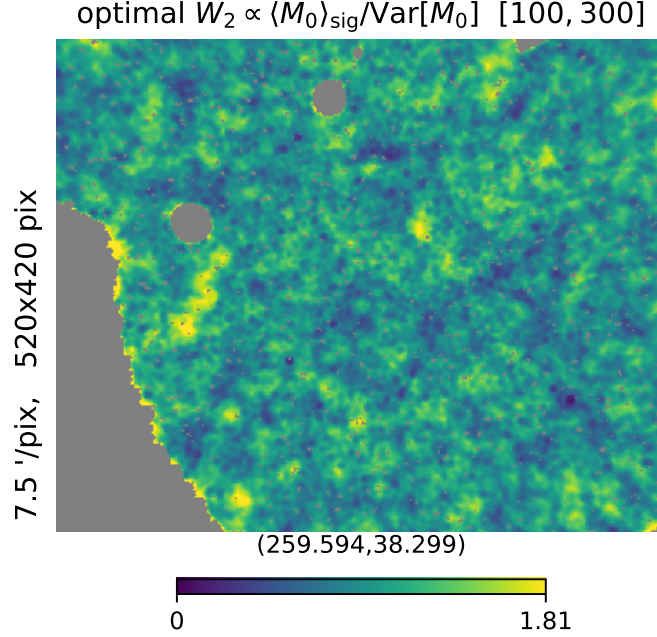


Figure 3: The optimal post-square weight $W_2 \propto \mathbb{E}[M_0]_{\text{sig}} / \text{Var}[M_0]$ for band [100, 300], built deterministically from the analytic signal mean and total variance (normalized to unit mean). Its non-trivial spatial structure shows that uniform post-square weighting is sub-optimal; the variation is set by the filter-smoothed depth, as in Eq. (29).